

STATIC ROUTING

Project 2

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A multi-router static routing lab: interface addressing, static route configuration across six routers, PC endpoint setup, and end-to-end connectivity verification.

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Overview

This project configures static routing across a six-router topology, connects three PCs to their respective gateway routers, and verifies end-to-end reachability by pinging between all three PC subnets.

Structure of this document:

- Task 1 — Creating static routes on Routers 1 through 6
- Task 2 — Assigning IP addresses to PCs 1, 2 and 3
- Task 3 — Verifying connectivity between all PCs via ping

Task 1: Creating Static Routes in the Router

Router 1 (R1)

Interface Configuration

- eth0/0 — 192.50.20.1
- eth0/1 — 12.0.0.1

Static Routes

Destination Network	Next Hop / Exit Interface
23.0.0.0/8	12.0.0.2
26.0.0.0/8	12.0.0.2
24.0.0.0/8	12.0.0.2
32.0.0.0/8	12.0.0.2
45.0.0.0/8	12.0.0.2
223.0.50.0/24	12.0.0.2
172.36.0.0/16	12.0.0.2

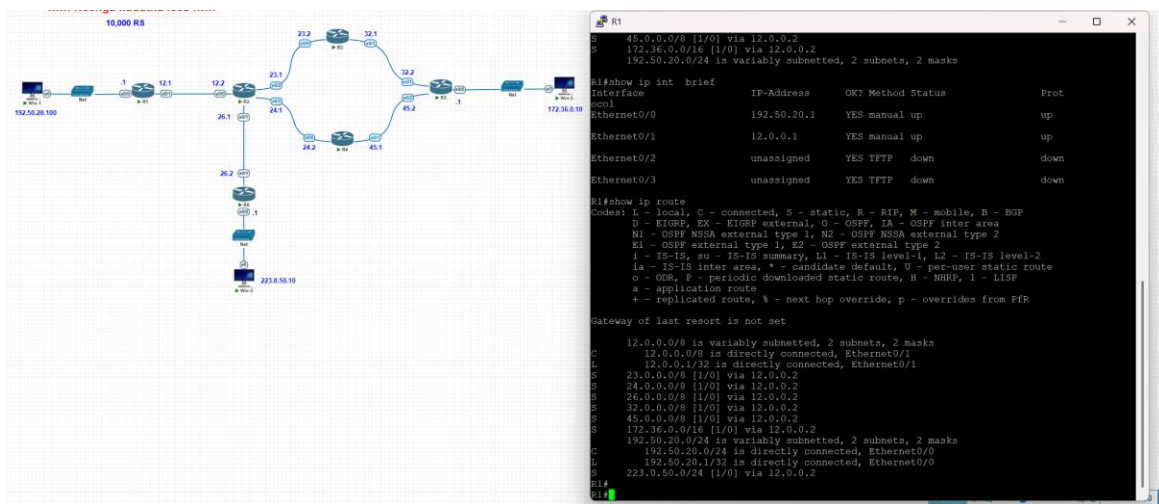


Figure 1.1 — Router 1 static route configuration

Router 2 (R2)

Interface Configuration

- eth0/0 — 12.0.0.2
- eth0/1 — 26.0.0.1
- eth0/2 — 23.0.0.1
- eth0/3 — 24.0.0.1

Static Routes

Destination Network	Next Hop / Exit Interface
32.0.0.0/8	23.0.0.2
45.0.0.0/8	24.0.0.2
223.0.50.0/24	26.0.0.2
172.36.0.0/16	23.0.0.2 & 24.0.0.2
192.50.20.0/8	12.0.0.1

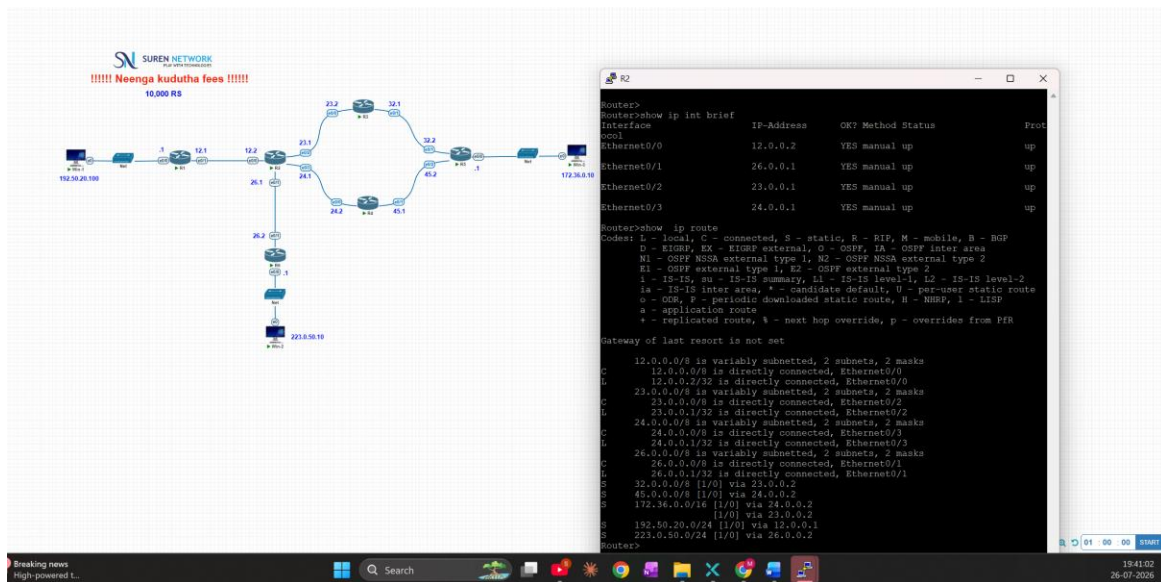


Figure 1.2 — Router 2 static route configuration

Router 3 (R3)

Interface Configuration

- eth0/0 — 23.0.0.2
- eth0/1 — 32.0.0.1

Static Routes

Destination Network	Next Hop / Exit Interface
12.0.0.0/8	23.0.0.1
192.50.20.0/8	23.0.0.1
223.0.50.0/24	23.0.0.1
172.36.0.0/16	32.0.0.2
26.0.0.0/8	23.0.0.1
24.0.0.0/8	23.0.0.1
45.0.0.0/8	32.0.0.2

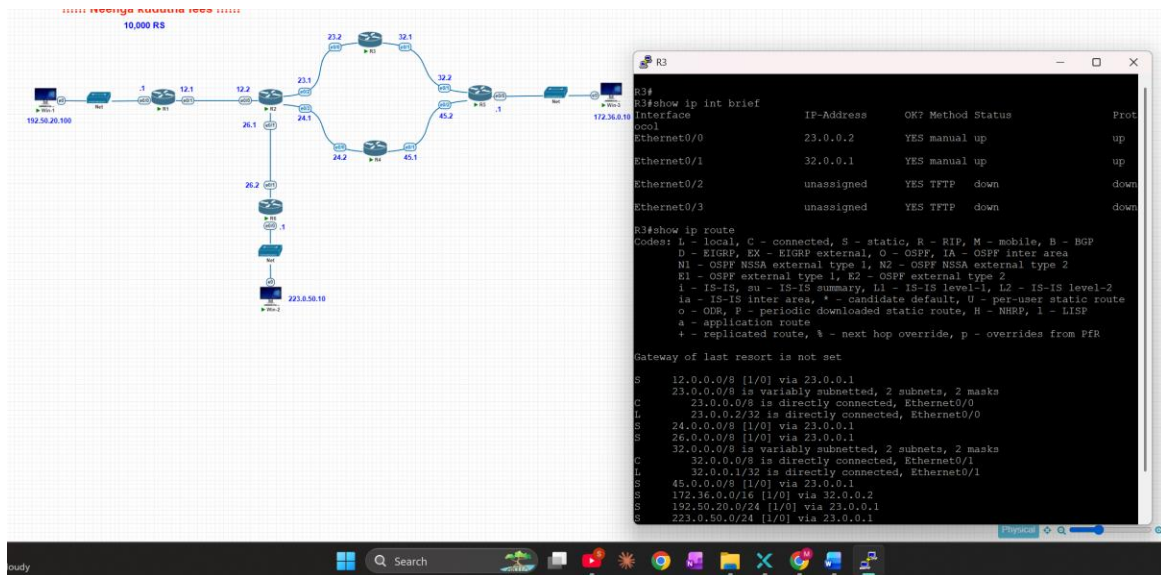


Figure 1.3 — Router 3 static route configuration

Router 4 (R4)

Interface Configuration

- eth0/0 — 24.0.0.2
- eth0/1 — 45.0.0.1

Static Routes

Destination Network	Next Hop / Exit Interface
12.0.0.0/8	24.0.0.1
192.50.20.0/8	24.0.0.1
223.0.50.0/24	24.0.0.1
172.36.0.0/16	45.0.0.2
26.0.0.0/8	24.0.0.1
23.0.0.0/8	24.0.0.1
32.0.0.0/8	45.0.0.2

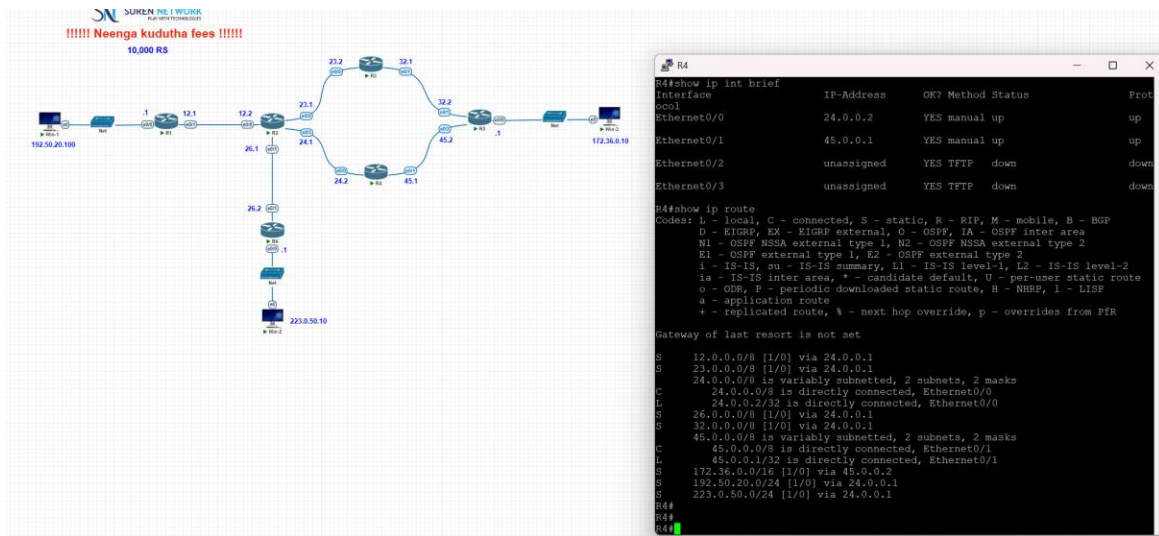


Figure 1.4 — Router 4 static route configuration

Router 5 (R5)

Interface Configuration

- eth0/0 — 172.36.0.10
- eth0/1 — 32.0.0.2
- eth0/2 — 45.0.0.2

Static Routes

Destination Network	Next Hop / Exit Interface
12.0.0.0/8	32.0.0.1 & 45.0.0.1
192.50.20.0/8	32.0.0.1 & 45.0.0.1
223.0.50.0/24	32.0.0.1 & 45.0.0.1
26.0.0.0/8	32.0.0.1 & 45.0.0.1
23.0.0.0/8	32.0.0.1
24.0.0.0/8	45.0.0.1

```

R5#
R5#show ip int brief
Interface                               IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0                             172.36.0.1      YES manual up          up
Ethernet0/1                             32.0.0.2        YES manual up          up
Ethernet0/2                             45.0.0.2        YES manual up          up
Ethernet0/3                             unassigned      YES TFTP  down        down

R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
      32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      32.0.0.0/8 is directly connected, Ethernet0/1
L      32.0.0.2/32 is directly connected, Ethernet0/1
      45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      45.0.0.0/8 is directly connected, Ethernet0/2
L      45.0.0.2/32 is directly connected, Ethernet0/2
      172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.36.0.0/16 is directly connected, Ethernet0/0
L      172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
R5#

```

Figure 1.5 — Router 5 static route configuration

Router 6 (R6)

Interface Configuration

- eth0/0 — 223.0.50.1
- eth0/1 — 26.0.0.2

Static Routes

Destination Network	Next Hop / Exit Interface
12.0.0.0/8	26.0.0.1

Destination Network	Next Hop / Exit Interface
192.50.20.0/8	26.0.0.1
172.36.0.0/16	26.0.0.1
23.0.0.0/8	26.0.0.1
32.0.0.0/8	26.0.0.1
45.0.0.0/8	26.0.0.1
24.0.0.0/8	26.0.0.1

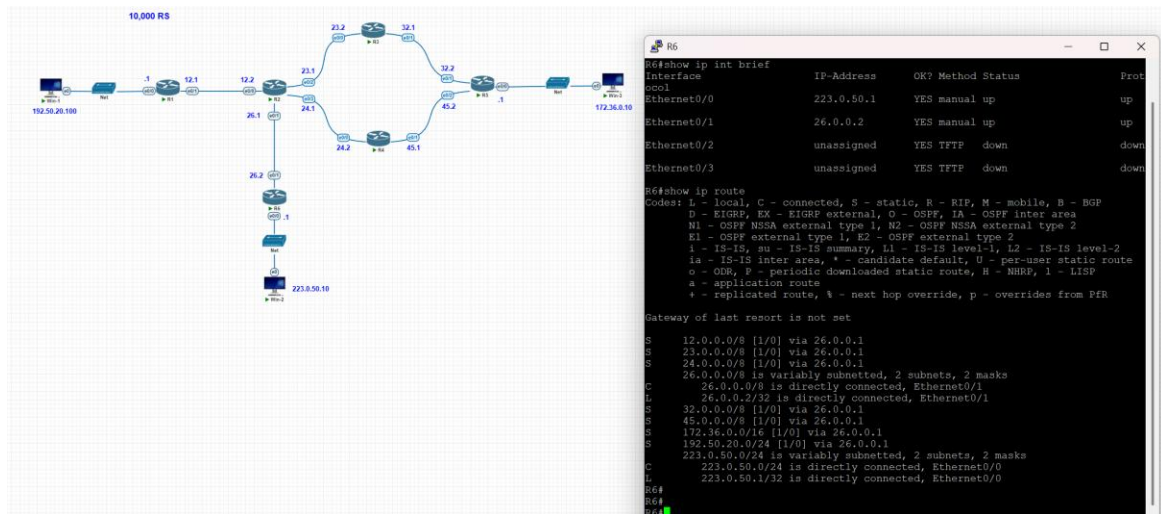


Figure 1.6 — Router 6 static route configuration

Task 2: Assigning IP Addresses to PCs

PC 1

IP Address: 192.50.20.100 Gateway: 192.50.20.1

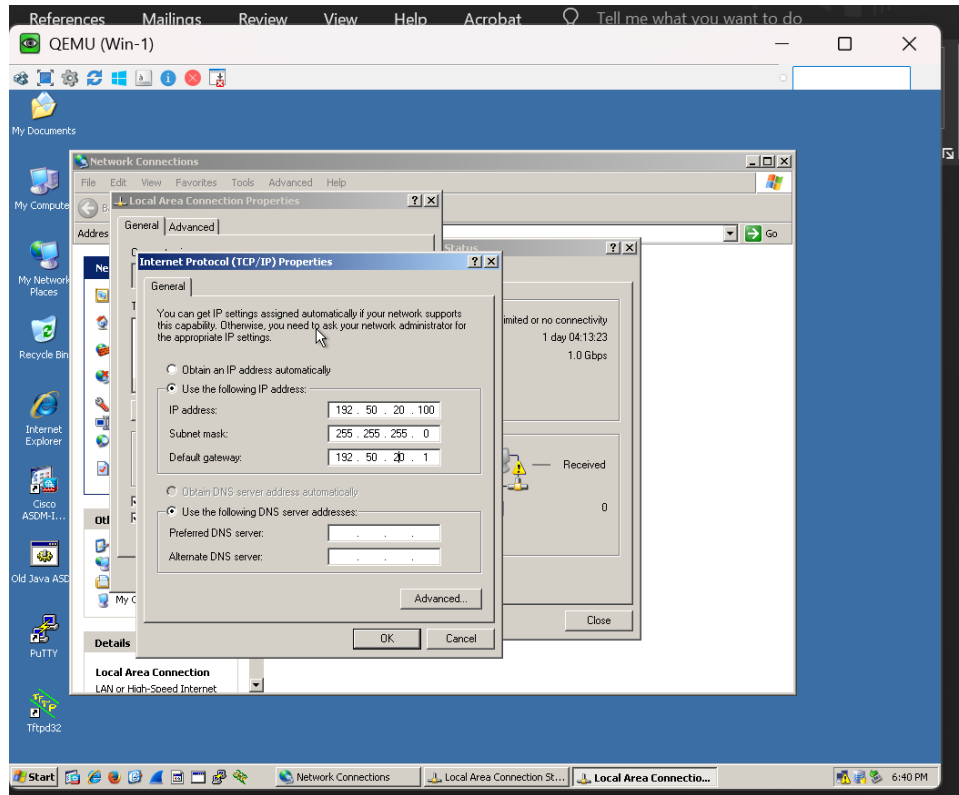


Figure 2.1 — PC 1 IP configuration

PC 2

IP Address: 223.0.50.10 Gateway: 223.0.50.1

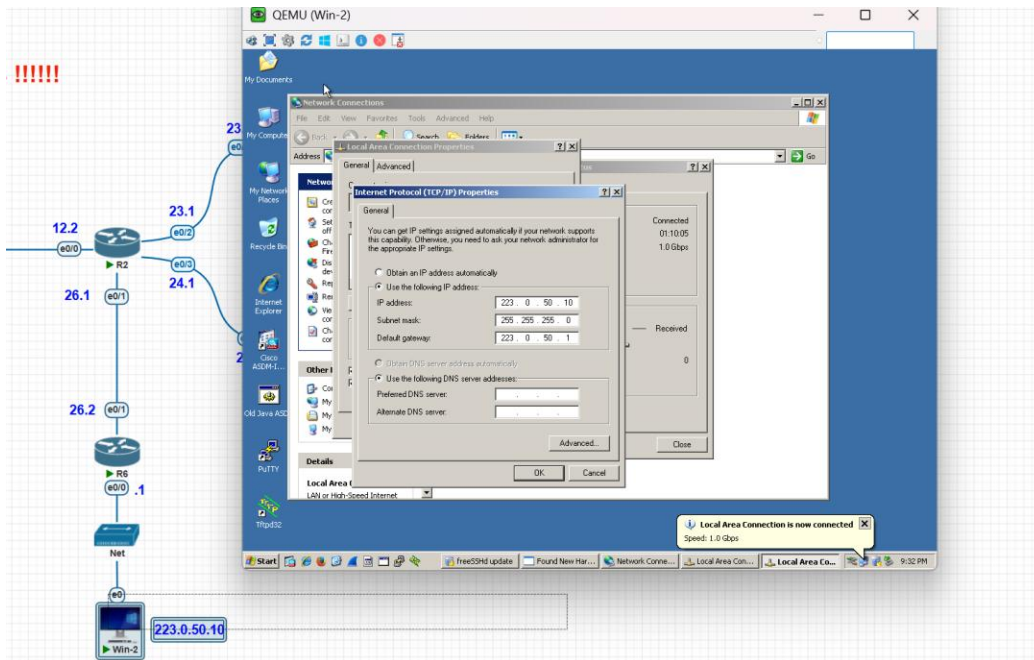


Figure 2.2 — PC 2 IP configuration

PC 3

IP Address: 172.36.0.10 Gateway: 172.36.0.1

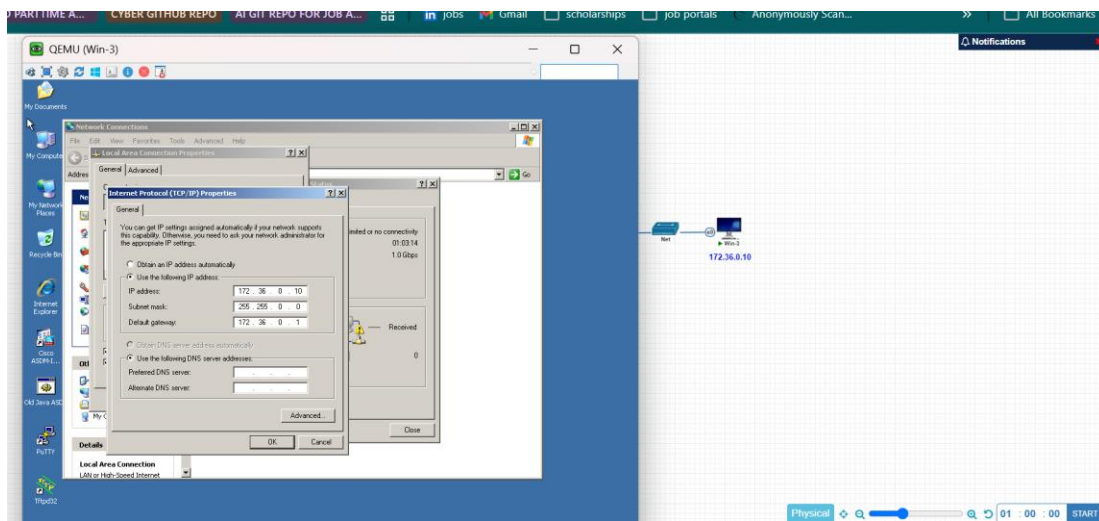


Figure 2.3 — PC 3 IP configuration

Task 3: Verifying the Connection Establishments

1. Ping from PC 1 (192.50.20.100)

- Ping from 192.50.20.100 (PC1) to 223.0.50.10 (PC2)
- Ping from 192.50.20.100 (PC1) to 172.36.0.10 (PC3)

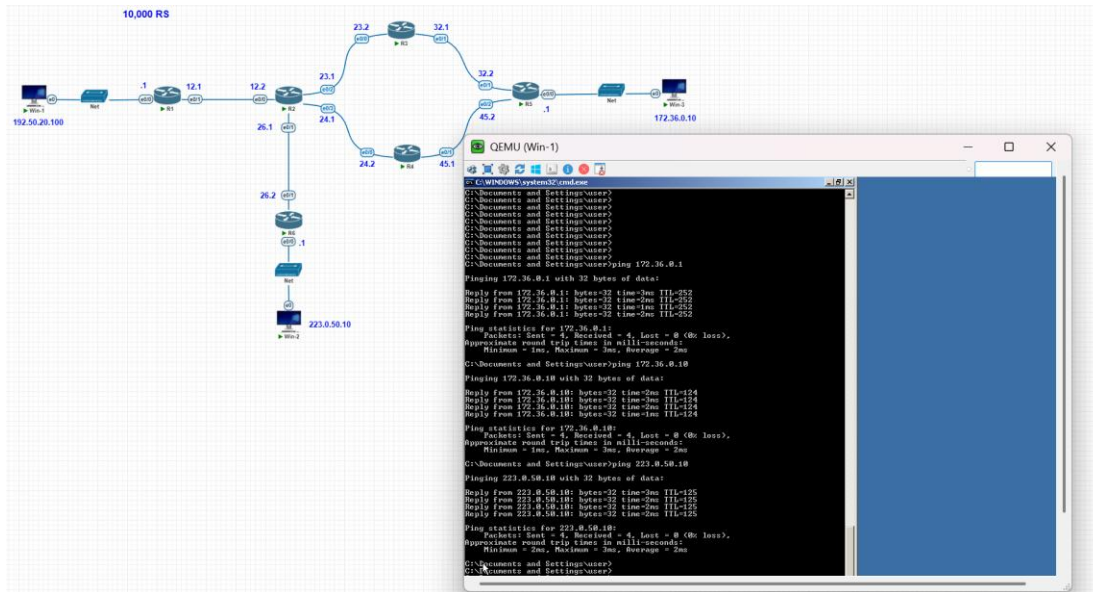


Figure 3.1 — Ping results from PC 1

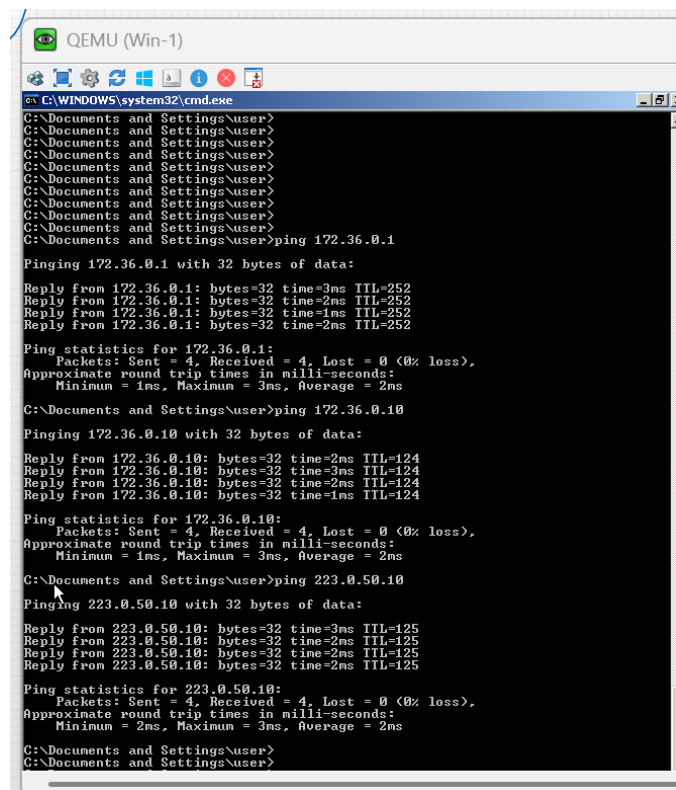
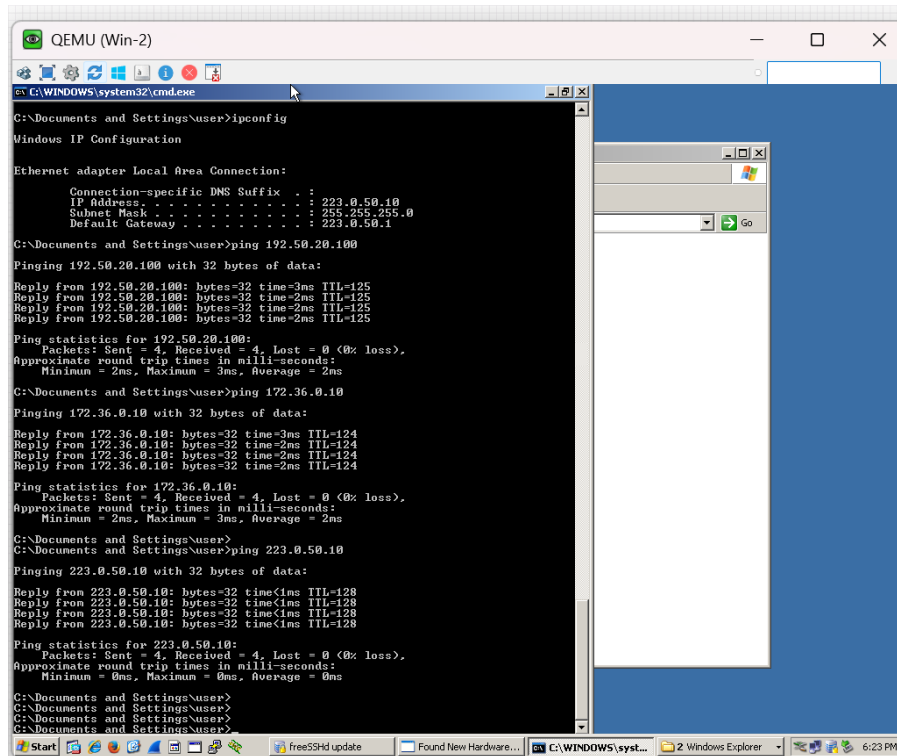


Figure 3.2 — Ping results from PC 1 (continued)

2. Ping from PC 2 (223.0.50.10)

- Ping from 223.0.50.10 (PC2) to 192.50.20.100 (PC1)
- Ping from 223.0.50.10 (PC2) to 172.36.0.10 (PC3)



```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 223.0.50.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 223.0.50.1

C:\Documents and Settings\User>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=3ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>
C:\Documents and Settings\User>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\User>
C:\Documents and Settings\User>
C:\Documents and Settings\User>
```

Figure 3.3 — Ping results from PC 2

3. Ping from PC 3 (172.36.0.10)

- Ping from 172.36.0.10 (PC3) to 192.50.20.100 (PC1)
- Ping from 172.36.0.10 (PC3) to 223.0.50.10 (PC2)

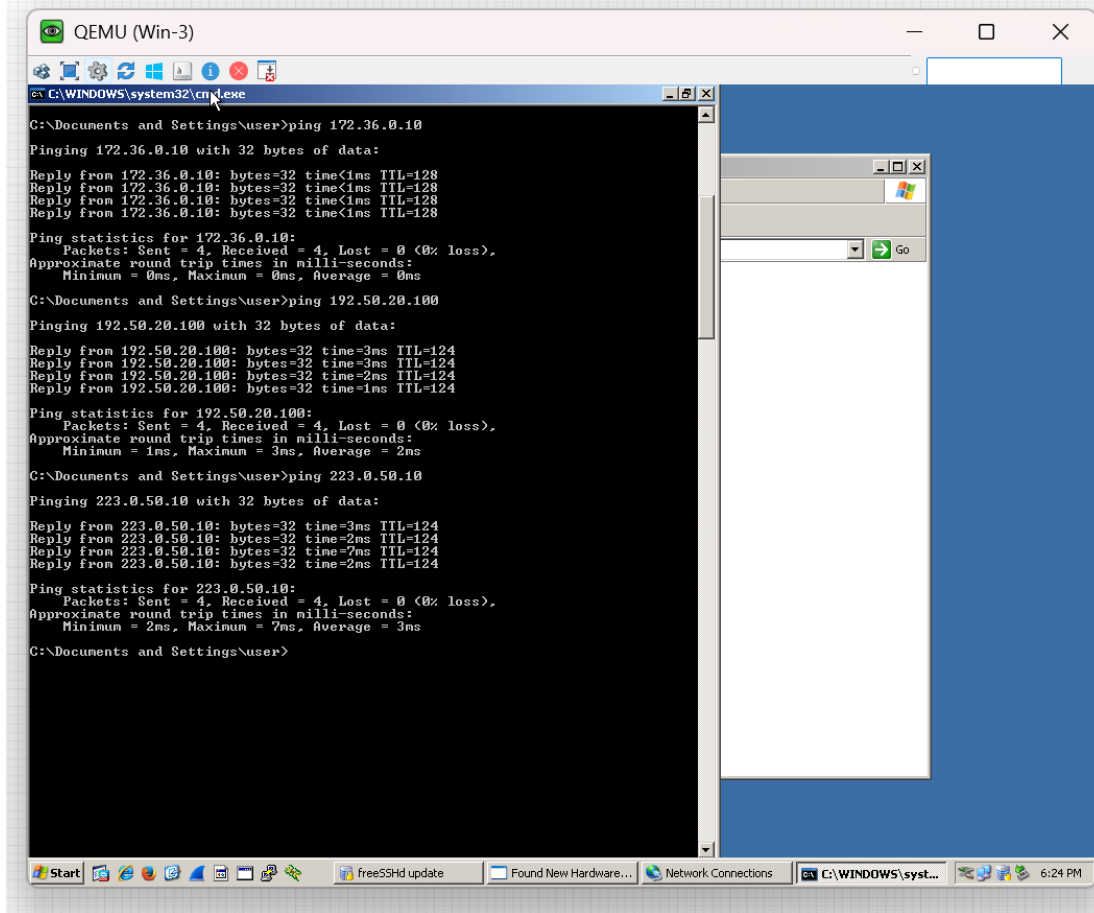


Figure 3.4 — Ping results from PC 3

Task 4: Identifying the Errors

Error 1: Destination Host Unreachable

There is no route for the destination IP in the source PC's route table.

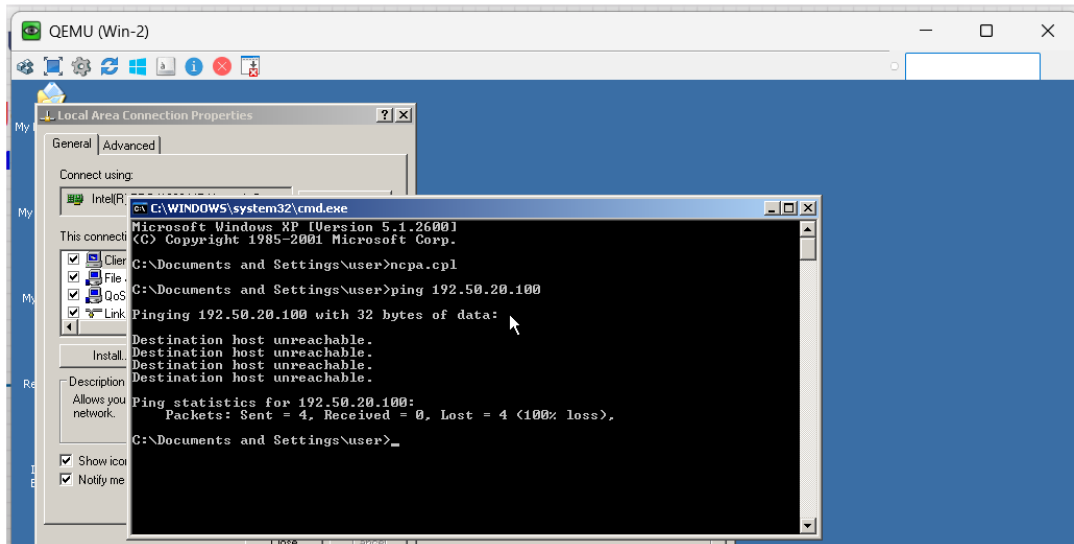


Figure 4.1 — Destination host unreachable

Error 2: Request Timed Out

The data packet was sent, but no reply was received within the expected time.

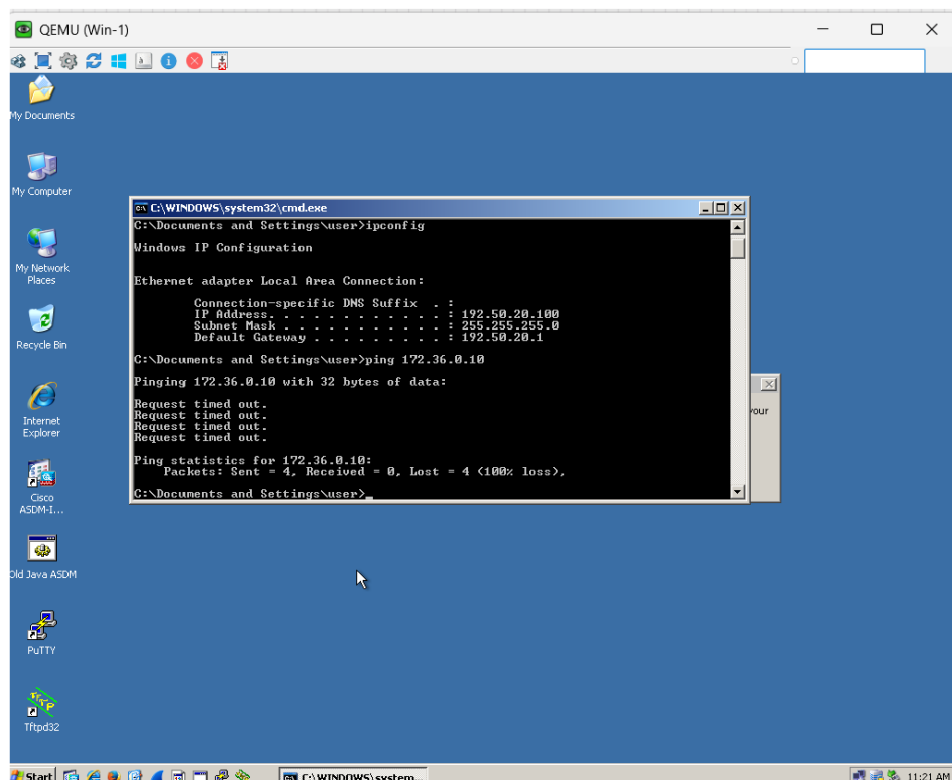


Figure 4.2 — Request timed out

Error 3: Reply from 192.50.20.1 — Destination Host Unreachable

The router (192.50.20.1) responded that it could not reach the destination host.

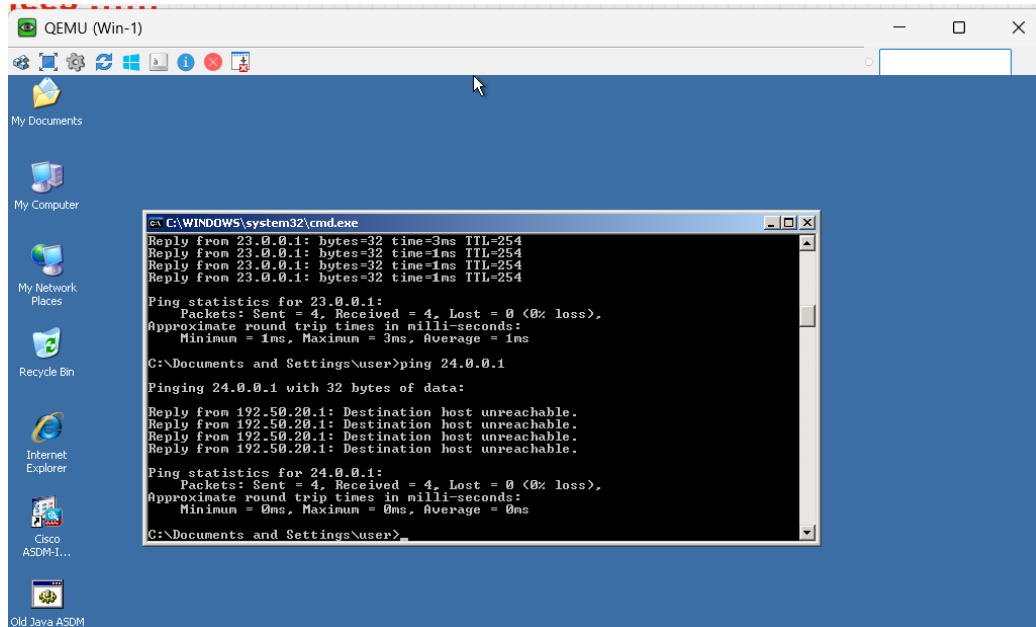


Figure 4.3 — Reply from 192.50.20.1, destination host unreachable

Error 4: Reply from 12.0.0.2 — TTL Expired in Transit

The packet exceeded its maximum hop limit (TTL) before reaching the destination and was discarded.

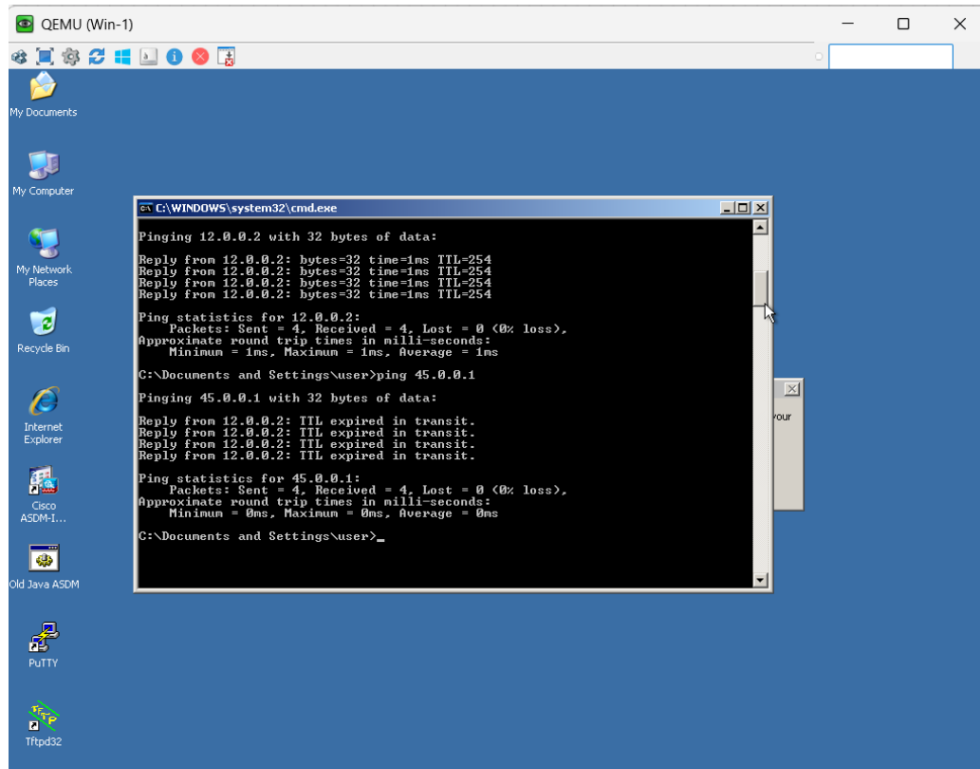


Figure 4.4 — Reply from 12.0.0.2, TTL expired in transit

How the TTL Expired Error Occurred

Router 1 Configuration

In R1, the IP route for 45.0.0.0 is 12.0.0.2 (R2).

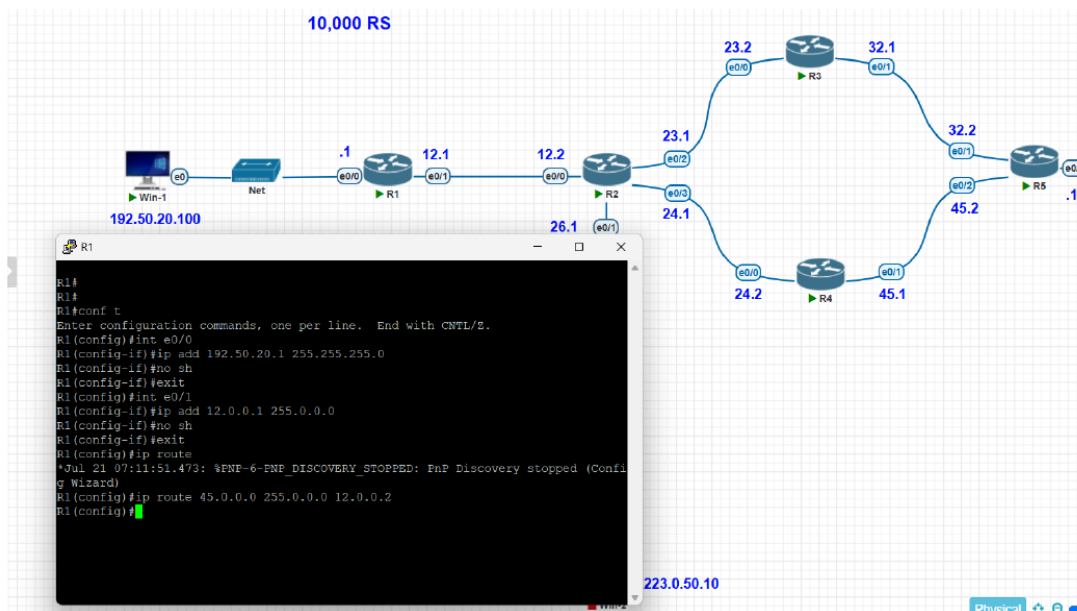


Figure 4.5 — Router 1 route for 45.0.0.0 via R2

In R2, the IP route for 45.0.0.0 is 12.0.0.1 (R1).



Because R1 points to R2 and R2 points back to R1 for the same destination (45.0.0.0), packets loop between the two routers, decrementing their TTL on each hop until it reaches zero — producing the "TTL expired in transit" error.

Conclusion

All three PCs (PC1, PC2, PC3) successfully reach one another across the six-router topology, confirming that static routes on every router correctly direct traffic toward each remote subnet and back. Task 4 further shows that a routing loop between R1 and R2 for the 45.0.0.0 network — caused by each router pointing to the other as the next hop — produces a TTL-expired-in-transit error, alongside other common ping failures (unreachable host, timeout) traceable to missing or incorrect route entries.